

```

# Initialize the board with numbers 1 through 9
board = [" " for _ in range(9)]

def print_board():
    # Displays the current 3x3 board state
    print()
    print(f" {board[0]} | {board[1]} | {board[2]} ")
    print("-----")
    print(f" {board[3]} | {board[4]} | {board[5]} ")
    print("-----")
    print(f" {board[6]} | {board[7]} | {board[8]} ")
    print()

def check_win(player):
    # Winning combinations (rows, columns, diagonals)
    win_conditions = [
        [0, 1, 2], [3, 4, 5], [6, 7, 8], # Rows
        [0, 3, 6], [1, 4, 7], [2, 5, 8], # Columns
        [0, 4, 8], [2, 4, 6]             # Diagonals
    ]
    for condition in win_conditions:
        if board[condition[0]] == board[condition[1]] == board[condition[2]] == player:
            return True
    return False

def check_tie():
    return " " not in board

def play_game():
    current_player = "X"
    print("Name=Jasti yashas chandrika   USN: 1BM24CS123")
    print("Welcome to Tic Tac Toe!")

    while True:
        print_board()
        print(f"Player {current_player}'s turn.")

        try:
            choice = int(input("Choose a position from 1 to 9: ")) - 1
        except ValueError:
            print("Invalid input! Please enter a number between 1 and 9.")
            continue

        if choice < 0 or choice > 8:
            print("Out of bounds! Choose a position between 1 and 9.")
            continue

        if board[choice] != " ":
            print("That space is already occupied! Try another one.")
            continue

        board[choice] = current_player

        if check_win(current_player):
            print_board()
            print(f"Player {current_player} wins!")
            break

        if check_tie():
            print_board()
            print("It's a tie!")
            break

        # Switch player
        current_player = "O" if current_player == "X" else "X"

if __name__ == "__main__":
    play_game()

```

Name=Jasti yashas chandrika USN: 1BM24CS123
Welcome to Tic Tac Toe!

```

| |
-----+-----
| |

```

```
---+---+---  
| | |
```

Player X's turn.

Choose a position from 1 to 9: 1

```
 X | |  
---+---+---  
| | |  
---+---+---  
| | |
```

Player O's turn.

Choose a position from 1 to 9: 2

```
 X | O |  
---+---+---  
| | |  
---+---+---  
| | |
```

Player X's turn.

Choose a position from 1 to 9: 3

```
 X | O | X  
---+---+---  
| | |  
---+---+---  
| | |
```

Player O's turn.

Choose a position from 1 to 9: 4

```
 X | O | X  
---+---+---  
 O | |  
---+---+---  
| | |
```

Player X's turn.

Choose a position from 1 to 9: 7

```
 X | O | X  
---+---+---  
 O | |  
---+---+---  
 X | |
```

Player O's turn.

Choose a position from 1 to 9: 4

That space is already occupied! Try another one.